

D - Card Pile Query

分值: 400 分

Problem Statement

There are N cards and N piles of cards.

有 N 张卡片和 N 堆卡片。

The cards and the piles are numbered $1, 2, \dots, N$. Initially, pile i contains only card i .

卡片和堆的编号为 $1, 2, \dots, N$ 。初始时, 堆 i 中只有卡片 i 。

Perform the following operation for each $i = 1, 2, \dots, Q$ in order:

按顺序对每个 $i = 1, 2, \dots, Q$ 执行以下操作:

- Move card C_i and all cards stacked on top of card C_i , preserving their order, on top of card P_i . It is guaranteed that immediately before performing the operation, cards C_i and P_i are in different piles, and card P_i is on top of some pile.
- 将卡片 C_i 以及堆在它上方的所有卡片保持原有顺序, 移动到卡片 P_i 的上方。题目保证在执行该操作前, 卡片 C_i 和 P_i 位于不同的堆中, 且卡片 P_i 处于某个堆的顶部。

Find the number of cards in each pile after all operations are completed.

求所有操作完成后每堆中的卡片数量。

Constraints

- $1 \leq N, Q \leq 3 \times 10^5$
- $1 \leq C_i, P_i \leq N$
- When the operations are performed in order, immediately before each operation, cards C_i and P_i are in different piles.
- When the operations are performed in order, immediately before each operation, card P_i is on top of some pile.

- All input values are integers.
- $1 \leq N, Q \leq 3 \times 10^5$
- $1 \leq C_i, P_i \leq N$
- 按顺序执行操作时，每次操作前，卡片 C_i 和 P_i 都位于不同的堆中。
- 按顺序执行操作时，每次操作前，卡片 P_i 都处于某个堆的顶部。
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
N Q
C1 P1
C2 P2
⋮
CQ PQ
```

Output

Let A_i be the number of cards in pile i at the end. Output $A_1, A_2, \dots, A_N$ in this order, separated by spaces.

设 A_i 为最终堆 i 中的卡片数量。按顺序输出 $A_1, A_2, \dots, A_N$ ，数值之间用空格分隔。

Sample Input 1

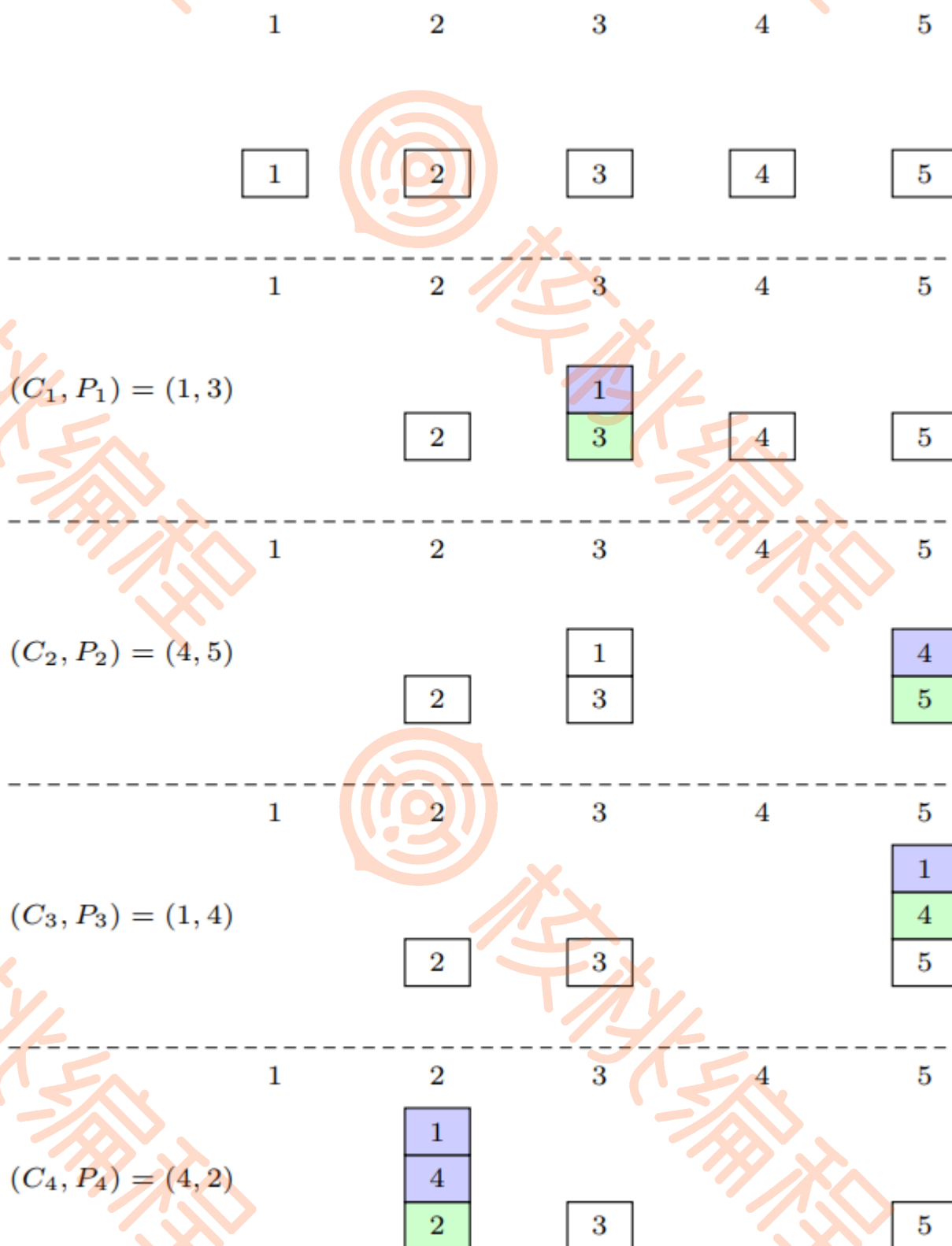
```
5 4
1 3
4 5
1 4
4 2
```

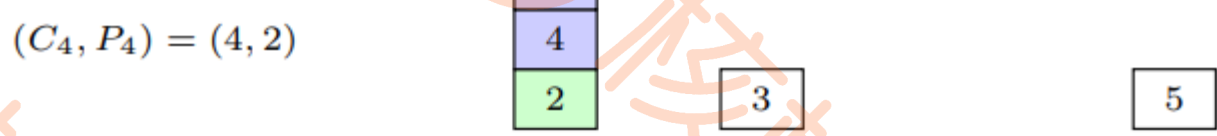
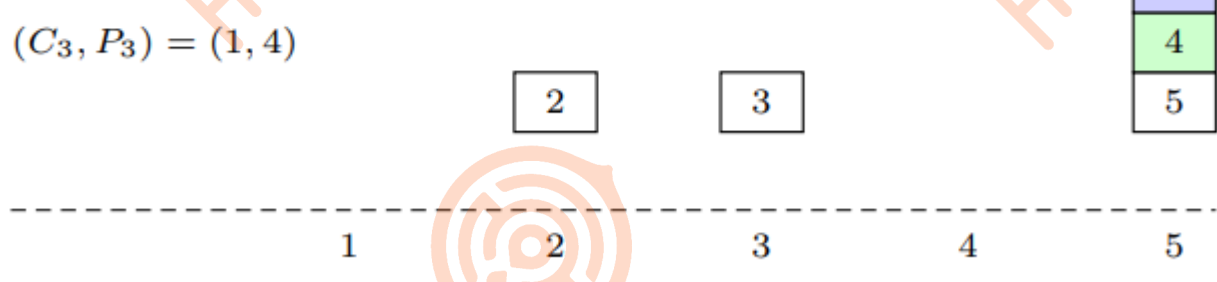
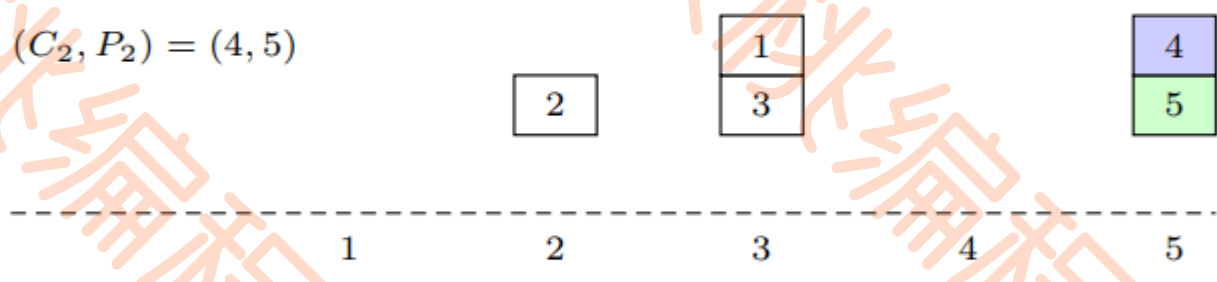
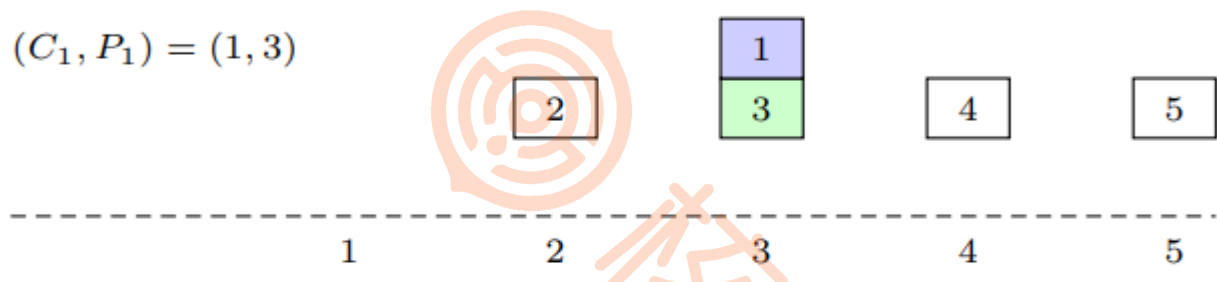
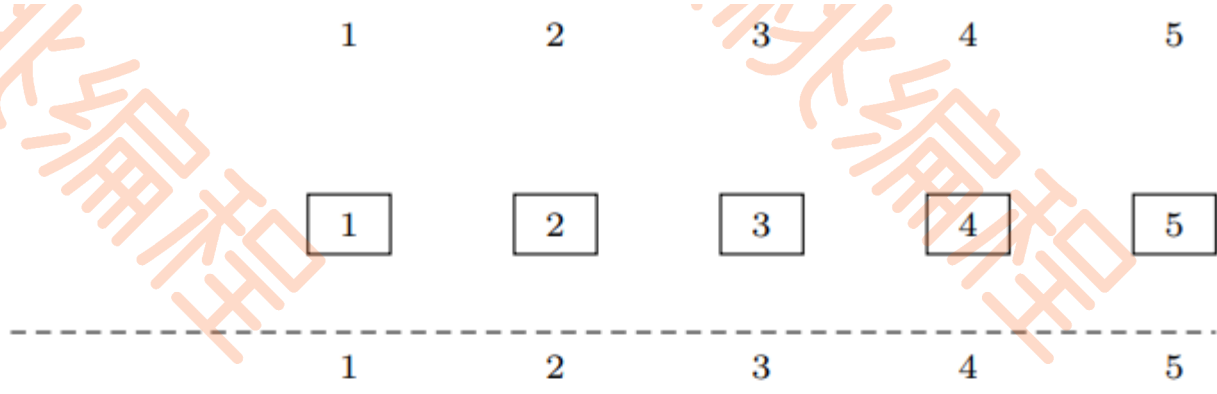
Sample Output 1

0 3 1 0 1

The sequence of operations proceeds as shown in the figure below.

操作序列的执行过程如下图所示。





Sample Input 2

```
7 8
3 1
5 4
```

```
2 5  
5 7  
2 3  
6 2  
3 4  
5 1
```

Sample Output 2

```
2 0 0 4 0 0 1
```